

Introduction*

Replication and Reanalysis

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Briefly describe the motivation, data, analysis, key findings, and implications. Clearly describe what the reader learns as a result of this paper. Three to five sentences.

1 Introduction

Clinical trials are the gold standard for evaluating treatment efficacy, but researchers often focus on treatment effects in the overall population when designing experiments. Accounting for the overall population creates a structural blind spot: demographic subgroups such as women, racial minorities, and ethnic minorities are often enrolled in numbers too small to support reliable inference about whether a treatment works equally well ??? or at all ??? for those groups. The absence of a significant finding is not evidence that the treatment effect is uniform when a trial lacks power to detect subgroup differences. Yet, many researchers and policymakers routinely make that interpretation and write guidelines accordingly.

The ACTG 320 trial (Hammer et al. 1997) is a landmark HIV clinical trial that provides an instructive case study for examining these limitations. Conducted in the mid-1990s, ACTG320 randomized 1,156 patients with advanced HIV disease to either a standard two-drug nucleoside regimen or a three-drug regimen adding the protease inhibitor indinavir. The trial successfully demonstrated that indinavir-containing therapy reduced the hazard of AIDS progression or death by approximately 50%??? a finding that transformed HIV treatment. The trial was powered for an overall treatment comparison; however, enrollment was highly imbalanced across demographic groups. Female participants comprised over 80% of the sample, and only a handful of participants and even fewer events represented several races/ethnicities. Whether indinavir's benefit was consistent across these groups was, by design, largely unanswerable.

The central research question of this paper is: given the observed effect sizes and event rates in ACTG 320, which demographic subgroups were adequately powered to detect treatment

*Project repository available at: <https://github.com/susanpeck/Math261BExperimentalDesignProject>

effects, and what enrollment would be required in a future study to achieve adequate power for those that were not? This question matters because the number of observed events, not enrollment counts alone, affects statistical power in survival analysis. A subgroup can be numerically represented in a trial while still contributing too few events to support stable hazard ratio estimation or meaningful hypothesis testing. Quantifying this gap is the first step toward designing trials that can actually answer subgroup-level questions.

This paper replicates the original ACTG 320 survival analysis using Cox proportional hazards models and Kaplan–Meier curves, then extends it through a systematic investigation of sex and race/ethnicity subgroup treatment effects and prospective power analyses. The key finding is that while the overall treatment effect and the female subgroup were adequately powered, the male, Asian/Pacific Islander, Hispanic, and Other race subgroups were not ???, and in some cases required dramatic increases in enrollment to achieve 80% power. The paper is organized as follows: Section 2 reviews the relevant literature on demographic representation in clinical trials; Section 3 describes the ACTG 320 study design; Section 4 presents the statistical analysis plan and power analysis framework; Section 5 presents the reanalysis results; and Section 6 discusses findings, limitations, and implications for future trial design.

2 Literature Review

The underrepresentation of women and racial and ethnic minorities in clinical trials has been documented across decades of pharmaceutical research, and three studies are particularly relevant to the questions examined here.

Poon et al. (2013) examined the participation of women and the reporting of sex-stratified analyses in 156 late-phase pivotal trials supporting FDA approval from 2000 to 2009. Women constituted, on average, 45% of enrolled participants, with the greatest imbalances in cardiovascular and infectious disease trials. Crucially, even when women were enrolled in adequate numbers, sex-specific analyses were rarely reported. Fewer than one-third of trials provided subgroup results stratified by sex. This finding maps directly onto the ACTG 320 sex subgroup problem: female participants dominated enrollment (82.6%) while the male stratum contributed only 15 events, leaving the sex-specific comparison severely underpowered regardless of total enrollment.

Khan et al. (2020) extended the question to racial and ethnic minorities, analyzing pivotal cardiometabolic drug trials submitted to the FDA between 2008 and 2017. Minority enrollment improved only modestly over the decade and remained well below the representation of those groups among patients with the diseases under study. The authors noted the same structural problem visible in ACTG 320: even when minority participants were enrolled, absolute numbers were too small to support meaningful subgroup analyses. They conclude by calling for prospective enrollment targets for underrepresented groups as a design requirement rather than a post-hoc consideration.

Eshera et al. (2015) documented the same pattern across all therapeutic areas in pivotal trials from 2005 to 2012, finding persistent underrepresentation of older adults, women, and racial minorities relative to disease burden in the general population. Taken together, these three studies establish that the limitations visible in ACTG 320 are not unique to HIV research but reflect a structural feature of how pivotal trials are designed. The absence of significant subgroup findings in these trials typically reflects insufficient power rather than an absence of heterogeneous treatment effects. A distinction that this paper quantifies explicitly using the ACTG 320 data.

3 Study Design

3.1 Experimental Design

The ACTG 320 trial (Hammer et al. 1997) was a multicenter, randomized, double-blind, placebo-controlled clinical trial evaluating whether adding the protease inhibitor indinavir to a standard two-drug nucleoside analogue regimen improved outcomes in HIV-infected patients with CD4 cell counts of 200 cells/mm³ or fewer. A total of 1,156 adults were enrolled across multiple clinical sites. Participants were randomly assigned, in approximately equal proportions, to control (two-drug regimen plus placebo) or treatment (two-drug regimen plus indinavir). Randomization was stratified by baseline CD4 count, separately for counts below 50 cells/mm³ and counts between 51 and 200 cells/mm³. Stratified randomization was the blocking mechanism that ensured balance on the most important prognostic variable across treatment arms. The double-blind, placebo-controlled structure prevented knowledge of treatment assignment from influencing outcome ascertainment.

3.2 Response Variable and Factors

The primary response variable is time-to-event: days from randomization to progression to an AIDS-defining illness or death, whichever occurred first. Participants who experienced neither outcome were censored at last contact. The primary factor of interest is treatment assignment (indinavir versus control). This paper additionally examines participant sex (female versus male) and race/ethnicity (White non-Hispanic, Black non-Hispanic, Hispanic, Asian/Pacific Islander, and Other) as potential effect modifiers. All models adjust for the following baseline covariates: age, CD4 cell count, Karnofsky performance score, months of prior ZDV use, and CD4 count stratum.

3.3 Strengths and Weaknesses

The design's strengths are substantial: randomization eliminates confounding in expectation, double-blinding prevents ascertainment bias, and stratified randomization ensures balance on

disease severity. The large overall enrollment provided strong power for the primary comparison, and the trial produced a landmark finding that transformed HIV treatment.

The primary weakness of this reanalysis is that the trial was not designed to support subgroup inference. The trial did not specify minimum subgroup sample sizes or event count targets for demographic groups. The result is a highly asymmetric event distribution. Female participants contribute 81 of the 96 observed events. Male participants contribute only 15, fewer than the minimum of 10-15 events per predictor typically required for stable Cox model estimation. The Asian/Pacific Islander and Other subgroups contribute 3 and 1 events, respectively, making subgroup-specific estimates for these groups entirely unreliable. Additionally, the publicly available dataset omits the trial's secondary endpoints and the American Indian/Alaska Native race category present in the original protocol, restricting this reanalysis to the primary outcome across five race/ethnicity groups.

4 Statistical Analysis Plan and Power Analysis

4.1 Dataset

The dataset used in this analysis is the ACTG 320 dataset, obtained from the `GLDreg` R library (Su 2023). The data contain participant-level records for 1,151 enrolled patients and were originally collected as part of a randomized clinical trial described in Hammer et al. (1997). The dataset was accessed as secondary data for this analysis; no new data were collected. The original trial enrolled 1,156 participants, and the five-record discrepancy between the published enrollment count and the available dataset is unexplained in the public data release.

The observational unit is the individual trial participant. Each record contains one row per participant with the following variables relevant to this analysis:

- **time**: days from randomization to the primary endpoint event or censoring
- **censor**: event indicator (1 = AIDS-defining event or death observed, 0 = censored)
- **tx**: treatment assignment (1 = indinavir regimen, 0 = control regimen)
- **sex**: participant sex (0 = female, 1 = male)
- **raceth**: race/ethnicity coded as a five-level factor (1 = White non-Hispanic, 2 = Black non-Hispanic, 3 = Hispanic, 4 = Asian/Pacific Islander, 5 = Other)
- **cd4**: baseline CD4 cell count (cells/mm³) measured at enrollment
- **age**: participant age in years at enrollment
- **karnof**: Karnofsky performance score, a clinical measure of functional status ranging from 0 to 100, with higher scores indicating better function
- **priorzdv**: months of prior zidovudine (ZDV) use before enrollment
- **strat2**: CD4 count stratum indicator used for stratified randomization (0 = CD4 < 50 cells/mm³, 1 = CD4 51–200 cells/mm³)

A total of 96 participants experienced the primary endpoint during follow-up, yielding an overall event rate of approximately 8.3%. The remaining 1,055 participants were censored, most at study close-out. The low overall event rate reflects the relatively short follow-up period of the trial. Moreover, statistical power for subgroup analyses is driven primarily by the small absolute number of events within each subgroup rather than by enrollment counts. The original article (Hammer et al. 1997) does not report a power analysis for subgroup comparisons; it only calculates power for the overall treatment effect.

Clinical trials evaluating time-to-event outcomes require statistical methods that properly account for censoring and the timing of events. In the context of the ACTG 320 study, the primary endpoint is the time to progression to AIDS or death, making survival analysis the appropriate framework. The section will start with a review of key concepts in survival analysis and progress to regression model formulations, hypothesis testing, and power analysis definitions.

4.2 Survival and Hazard Functions

Let T denote a non-negative random variable representing the time to the event of interest. The survival function is defined as

$$S(t) = P(T > t).$$

This quantity represents the probability that an individual remains event-free beyond time t and provides the basis for survival and hazard analysis. The hazard function is defined as

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T < t + \Delta t \mid T > t)}{\Delta t} \quad (1).$$

where Δt is a small time interval, and $h(t)$ represents the instantaneous event rate at time t conditional on survival up to time t (Moore 2016a). The hazard function represents how quickly the survival probability declines over time. Higher values of $h(t)$ correspond to a greater risk of the event occurring immediately after time t , while lower values indicate slower event occurrence. The hazard function can be derived from equation (1) as the ratio of the pdf of T and the survival function

$$h(t) = \frac{f(t)}{S(t)}.$$

4.3 Cox Proportional Hazards Model and Hazard Ratios

The Cox proportional hazards model estimate the association between covariates and the event outcomes in time-to-event clinical studies. The model is semi-parametric since it doesn't require specifying the baseline hazard function. For subject i with covariate vector X_i , the Cox model is

$$h_i(t | \mathbf{x}_i) = h_0(t) \exp(\mathbf{x}_i^\top \beta),$$

where $h_0(t)$ is the unspecified baseline hazard function and β is a vector of regression coefficients. In the Cox model, comparisons between groups are made through ratios of hazard functions, allowing treatment effects to be interpreted as relative differences in instantaneous risk over time (Moore 2016a). The hazard ratio (HR) compares the hazard function between two groups

$$HR = \frac{h_T(t)}{h_C(t)}$$

where $h_T(t)$ is the hazard function for the treatment group and $h_C(t)$ hazard function for the control group. An HR less than one indicates the treatment reduces the instantaneous risk and is more effective than the control group while an HR greater than one indicates the converse a detrimental treatment effect. Since the hazard ratios don't depend on t , the Cox model assumes a constant hazard ratio over all time intervals which can be interpreted that the proportion between treatment and control hazard values remains constant.

The Cox proportional hazards model is estimated using partial likelihood estimation to account for censoring of the data (Moore 2016b),

$$L(\beta) = \prod_{i:\delta_i=1} \frac{\exp(X_i^\top \beta)}{\sum_{j \in R(t_i)} \exp(X_j^\top \beta)}.$$

The partial likelihood is converted to the log-likelihood which is maximized to obtain the estimate $\hat{\beta}$.

4.4 Hypothesis Testing for Subgroup Interactions

The likelihood ratio test is generally preferred in survival analysis when evaluating potential changes in interaction effects (Moore 2016b). Let $\ell(\beta)$ denote the log partial likelihood of the Cox model. To test a hypothesis about a subset of parameters, two models are considered:

- A **reduced model**, which imposes the null hypothesis constraints, and

- A **full model**, which includes all parameters of interest.

The likelihood ratio test (LRT) statistic is defined as

$$\Lambda = -2 \left[\ell(\hat{\beta}_{\text{reduced}}) - \ell(\hat{\beta}_{\text{full}}) \right].$$

Assuming the null hypothesis is true, the test statistic follows an approximate chi-square distribution:

$$\Lambda \sim \chi_k^2,$$

where k is the number of restrictions imposed under the null hypothesis. For example, $k = 1$ for a single interaction term and $k > 1$ for joint tests involving multiple subgroup effects. The null hypothesis is rejected at significance level α if

$$\Lambda > \chi_{k, 1-\alpha}^2.$$

Primary Treatment Effect Model

The overall treatment effect may be evaluated using a Cox proportional hazards model of the form

$$h_i(t) = h_0(t) \exp \left(\beta_1 T_i + \beta_2^\top Z_i \right)$$

where T_i is the treatment indicator and Z_i is a vector of additional baseline covariates. The coefficient β_1 represents the log hazard ratio for the primary treatment effect after adjusting for the covariates in Z_i . The treatment hazard ratio is

$$HR_T = \exp(\beta_1)$$

The null hypothesis for the overall treatment effect is

$$\begin{aligned} H_0 : \beta_1 = 0 \quad \text{vs.} \quad H_A : \beta_1 \neq 0 \\ \text{or equivalently} \\ H_0 : HR_T = 1 \quad \text{vs.} \quad H_A : HR_T \neq 1 \end{aligned}$$

4.5 Measuring Subgroup Treatment Effects Between Sex

To evaluate whether the treatment effect differs by sex, the regression model is expanded to include a gender indicator and a treatment-by-gender interaction term:

$$h_i(t) = h_0(t) \exp \left(\beta_1 T_i + \beta_2 G_i + \beta_3 T_i G_i + \beta_4^\top Z_i \right)$$

where G_i is a binary gender indicator and Z_i represents additional covariates, β_1 is the treatment effect for the reference gender group, β_2 is the baseline difference in hazard between gender groups among participants in the control treatment group, and β_3 is the treatment-by-gender interaction effect.

The hypothesis test for whether the treatment effect differs by gender is

$$\begin{aligned} H_0 : \beta_3 = 0 \quad \text{vs.} \quad H_A : \beta_3 \neq 0 \\ \text{or equivalently} \\ H_0 : \exp(\beta_3) = 1 \quad \text{vs.} \quad H_A : \exp(\beta_3) \neq 1 \end{aligned}$$

4.6 Measuring Subgroup Treatment Effects Between Ethnic Groups

The same interaction framework can be extended to evaluate whether treatment effects differ across ethnic groups. In the ACTG320 trial, ethnicity is coded as: 1 = White non-Hispanic, 2 = Black non-Hispanic, 3 = Hispanic, 4 = Asian or Pacific Islander, and 6 = Other or unknown. (The category 5 = American Indian or Alaska Native is not present in the analysis dataset.)

Using White non-Hispanic participants as the reference group, the model is defined as

$$\begin{aligned} h_i(t) = h_0(t) \exp \left(\beta_1 T_i + \beta_2 Black_i + \beta_3 Hispanic_i + \beta_4 AsianPI_i + \beta_5 Other_i \right. \\ \left. + \beta_6 T_i Black_i + \beta_7 T_i Hispanic_i + \beta_8 T_i AsianPI_i + \beta_9 T_i Other_i + \beta_{10}^\top Z_i \right) \end{aligned}$$

In this model, β_1 is the treatment effect for White non-Hispanic participants. The coefficients β_2 through β_5 represent baseline hazard differences between each ethnic subgroup and the White non-Hispanic reference group among participants in the control treatment group.

The interaction coefficients β_6 through β_9 measure whether the treatment effect differs between each ethnic subgroup and the White non-Hispanic reference group. The vector Z_i includes additional baseline covariates.

The joint hypothesis test for whether treatment effects differ across ethnic groups is

$H_0 : \beta_6 = \beta_7 = \beta_8 = \beta_9 = 0$ H_A : At least one interaction coefficient is not equal to zero

If the null hypothesis is rejected, there is evidence that the treatment effect differs for at least one ethnic subgroup relative to the White non-Hispanic reference group.

4.7 Power for the Overall Treatment Effect

For a two-group survival comparison, power can be estimated directly when the number of observed or anticipated events is known. This formulation is useful for sensitivity analysis because the available event count may be fixed by the completed study or by the expected duration of follow-up.

For a two-sided test at significance level α , the approximate power is

$$\text{Power} = \Phi \left(\sqrt{Dp(1-p)} |\log(HR)| - z_{1-\alpha/2} \right),$$

where D is the total number of observed or anticipated events, p is the proportion assigned to the treatment group, $1-p$ is the proportion assigned to the control group, HR is the target hazard ratio comparing treatment with control, $z_{1-\alpha/2}$ is the standard normal critical value for a two-sided test, and $\Phi(\cdot)$ is the standard normal cumulative distribution function (Schoenfeld 1981).

4.8 Power for Subgroup Differences: Sex

Since interaction effects are estimated from contrasts between subgroup-specific treatment effects, the effective amount of information is reduced relative to the overall treatment comparison. Consequently, substantially larger event counts are typically required to detect treatment effect differences across subgroups (Schoenfeld 1981).

For a two-sided interaction test at significance level α , the approximate power is

$$\text{Power} = \Phi \left(\sqrt{Dp_T(1-p_T)p_G(1-p_G)} |\log(HR_{\text{int}})| - z_{1-\alpha/2} \right),$$

where D is the total number of observed or anticipated events, p_T is the proportion assigned to the treatment group, p_G is the proportion of participants in one of the subgroup categories

(for example, one gender subgroup), HR_{int} is the interaction hazard ratio, $z_{1-\alpha/2}$ is the standard normal critical value for a two-sided test, and $\Phi(\cdot)$ is the standard normal cumulative distribution function (Schoenfeld 1981).

The interaction hazard ratio quantifies the relative difference between subgroup-specific treatment effects. For example,

$$HR_{\text{int}} = \frac{HR_1}{HR_0}.$$

4.9 Subgroup Power Analysis: Ethnicity

The same framework extends to ethnicity subgroups, substituting the ethnic subgroup proportion p_E for the sex subgroup proportion p_G . For a two-sided interaction test at significance level α , the approximate power is

$$\text{Power} = \Phi \left(\sqrt{D p_T (1 - p_T) p_E (1 - p_E)} |\log(HR_{\text{int}})| - z_{1-\alpha/2} \right),$$

p_T is the proportion assigned to the treatment group, p_E is the proportion of participants in one of the ethnic subgroup categories, HR_{int} is the interaction hazard ratio comparing subgroup-specific treatment effects, $z_{1-\alpha/2}$ is the standard normal critical value for a two-sided test, and $\Phi(\cdot)$ is the standard normal cumulative distribution function (Schoenfeld 1981).

5 Reanalysis

The analysis is organized into four sections: whole group, sex subgroup, race/ethnicity subgroup, and power analysis. We use R (R Core Team 2026) and the package “survival” (Therneau 2026) to perform the analysis and generate exhibits in this paper

5.1 Whole Group

The Kaplan-Meier curves for the control and treatment groups are shown in Figure 1 and visually suggest a noticeable reduction in the hazard.

The baseline Cox model includes the following covariates: treatment indicator (tx), baseline CD4 count (cd4), age, Karnofsky performance scale (karnof), months of prior ZDV use (priorzdv), and CD40 cell count stratification (strat2). The regression results are in Table 1

An $HR = 0.517$ for the treatment indicator means that at any given point in time, patients receiving indinavir had about half the hazard of experiencing an AIDS-defining event or death compared to patients not receiving indinavir after controlling for the other covariates, which is

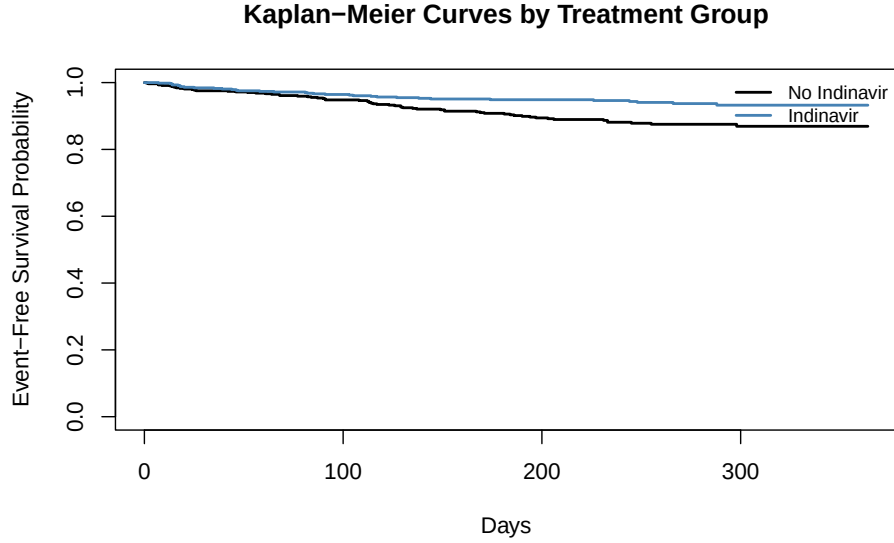


Figure 1: Kaplan-Meier survival curves for Indinavir versus No Indinavir treatment groups.

consistent with what Hammer et al. (1997) reported and confirms your model is reproducing the original study findings.

Table 1: Whole-Group Baseline Cox Model — Hazard Ratios

Covariate	HR	95% CI		p-value
		Lower	Upper	
Treatment: Indinavir vs. No Indinavir	0.517	0.339	0.788	0.002
Age	1.022	1.000	1.045	0.054
Baseline CD4	0.985	0.978	0.993	0.000
Karnofsky Score	0.948	0.926	0.970	0.000
Prior ZDV Use	1.000	0.992	1.007	0.925
CD4 Stratum	1.012	0.517	1.984	0.971

We used a log-rank test to test overall model significance. Based on the chi-square statistic of 10.545 with 1 degree of freedom, we obtained a p-value of $p = 0.0012$. This is statistically significant, indicating that patients receiving indinavir had meaningfully different event timing than those who did not.

5.2 Gender Subgroup

We fit two models to test the difference in treatment effects between men and women. The first is the baseline model, which adds `sex` as a fixed covariate alongside treatment. The second is the full model, which adds a `tx * sex` interaction term to test whether sex modifies the treatment effect. A likelihood ratio test compares the two models. Then, we fit two Cox models within each sex stratum (female-only, male-only) to calculate stratum-specific HRs.

5.2.1 Gender Baseline Model

Sex is included as a covariate with no interaction, estimating the marginal treatment effect after adjusting for sex. The results in Table 2 show the HR remains unchanged at 0.517. The HR for sex has an insignificant p-value, indicating that sex has no meaningful impact on the hazard ratio.

Table 2: Sex Baseline Cox Model — Hazard Ratios

Covariate	HR	95% CI Lower	95% CI Upper	p-value
Treatment (IDV vs no IDV)	0.517	0.339	0.789	0.002
Sex (Male vs Female)	1.122	0.643	1.957	0.685
Age	1.023	1.000	1.046	0.050
Baseline CD4	0.985	0.978	0.993	0.000
Karnofsky Score	0.948	0.926	0.970	0.000
Prior ZDV Use	1.000	0.992	1.007	0.926
CD4 Stratum	1.013	0.517	1.985	0.970

5.2.2 Sex Full Model (Interaction)

In the full model, the HR for treatment among females (the reference group for sex) is 0.460. This means females receiving indinavir have about 54% lower hazard than females not receiving it, a statistically significant difference. The interaction HR is 2.034, indicating the treatment effect in male participants appears weaker than that in female participants. However, the 95% CI is very wide and crosses 1.0, and $p = 0.2141$, so this difference is not statistically significant. We cannot conclude from this data that sex genuinely modifies the treatment effect.

Table 3: Sex Full Cox Model with Interaction — Hazard Ratios

Covariate	HR	95% CI Lower	95% CI Upper	p-value
Treatment (IDV vs no IDV)	0.460	0.288	0.734	0.001
Sex (Male vs Female)	0.854	0.405	1.801	0.679

Covariate	HR	95% CI Lower	95% CI Upper	p-value
Age	1.023	1.000	1.046	0.050
Baseline CD4	0.985	0.978	0.993	0.000
Karnofsky Score	0.947	0.925	0.969	0.000
Prior ZDV Use	0.999	0.992	1.007	0.865
CD4 Stratum	1.011	0.516	1.978	0.975
Treatment $\times$ Sex	2.034	0.663	6.238	0.214

The likelihood ratio test confirms this with a p-value = 0.2187. Adding the interaction term does not significantly improve model fit over the baseline model. We fail to reject the null hypothesis that the treatment effect is meaningfully different between female and male patients. The effect sizes by sex are estimated and shown below.

With only 15 events in the male stratum, the Cox model has very little information to work with. The model is severely underpowered, and the estimates are unreliable.

5.3 Race/Ethnicity Subgroup

Following the same procedure as with sex subgroups, we fit and compare a baseline model with no interactions and a full model with interactions. In this case, we include race/ethnicity as a 5-level factor with White as the reference group.

5.3.1 Baseline Race/Ethnicity Model

The treatment HR = 0.526 means that after adjusting for race/ethnicity as a 5-level factor, patients receiving indinavir had approximately 47% lower hazard of AIDS-defining events or death compared to patients not receiving indinavir. This result suggests that race/ethnicity is not a meaningful confounder of the treatment-outcome relationship, like the participant's gender.

5.3.2 Race Full Model

5.3.3 Race Full Model

The treatment HR for White patients (the reference group) is 0.534, confirming a significant protective effect.

The interaction terms for Black, Hispanic, and Other are all insignificant, suggesting no meaningful difference in treatment effect by race. The interaction for Asian/PI appears significant,

with a p-value = 00.033. However, the bounds of the 95% CI suggest highly unstable estimates due to an extremely small sample size.

We confirm these results with the likelihood ratio test with Chi-square = 8.326, df = 4, p = 0.080. Using the conventional significance of $\alpha = 0.05$, we conclude the interaction as a whole is not statistically significant. We cannot conclude that race/ethnicity modifies the treatment effect. The unstable Asian/PI interaction term is likely the reason for the smaller p-value. Excluding these observations from the data would likely increase the p-value.

The race/ethnicity subgroup analysis is limited by severe sample size constraints in the non-White groups, particularly Asian/PI and Other. The formally correct conclusion is that race/ethnicity does not significantly modify the treatment effect at $\alpha = 0.05$.

Table 4: Race Full Cox Model with Interaction — Hazard Ratios

Covariate	HR	95% CI Lower	95% CI Upper	p-value
Treatment (IDV vs no IDV)	0.534	0.302	0.946	0.032
Race: Black (vs White)	0.794	0.432	1.457	0.456
Race: Hispanic (vs White)	1.002	0.524	1.914	0.995
Race: Asian/PI (vs White)	0.860	0.117	6.317	0.882
Race: Other (vs White)	2.831	0.378	21.227	0.311
Age	1.024	1.001	1.048	0.038
Baseline CD4	0.985	0.978	0.993	0.000
Karnofsky Score	0.948	0.927	0.971	0.000
Prior ZDV Use	0.997	0.989	1.005	0.467
CD4 Stratum	1.010	0.513	1.986	0.978
Treatment $\times$ Black	0.591	0.185	1.883	0.373
Treatment $\times$ Hispanic	1.249	0.422	3.698	0.687
Treatment $\times$ Asian/PI	15.514	1.244	193.477	0.033
Treatment $\times$ Other	0.000	0.000	Inf	0.993

Table 4 contains the estimated race specific effect sizes.

5.4 Summary of All Effect Sizes

Table 5 summarizes all effect sizes as measured by HR. They are stable for the overall group, sex, and race, but become unreliable at the subgroup level due to the small number of events. For Asian/PI and Other, the estimates are completely unreliable.

HR < 1 indicates indinavir-containing regimens reduce the hazard of AIDS-defining events or death relative to nucleoside-only therapy. Interaction p-values from the LR tests indicate whether subgroup membership significantly modifies the treatment effect. Note that the

Asian/PI and Other race subgroups have small sample sizes, resulting in meaningless confidence intervals.

Table 5: Summary of Treatment Effect Sizes Across All Analyses

Analysis	Model	Subgroup	n	Events	HR	95% CI Lower	95% CI Upper
Whole Group	Baseline	All	1151	96	0.517	0.339	0.788
Sex	Baseline (pooled)	All	1151	96	0.517	0.339	0.789
Sex	Stratum-specific	Female	951	81	0.459	0.288	0.732
Sex	Stratum-specific	Male	200	15	1.101	0.393	3.087
Race	Baseline (pooled)	All	1151	96	0.526	0.344	0.805
Race	Stratum-specific	White	596	51	0.539	0.303	0.960
Race	Stratum-specific	Black	327	21	0.287	0.104	0.794
Race	Stratum-specific	Hispanic	203	20	0.628	0.246	1.602
Race	Stratum-specific	Asian/PI	14	3	0.068	0.000	Inf
Race	Stratum-specific	Other	11	1	237.311	0.000	Inf

5.5 Initial Observed Power for ACTG 320 Subgroup Analyses

Before estimating the sample sizes needed for future studies, we calculated the observed or effective power of the ACTG 320 analyses for the overall study population and for each sex and race/ethnicity subgroup.

Power was estimated assuming the treatment and control groups have equal sample sizes. We used $\alpha = 0.025$ to account for the Bonferroni-adjusted two-sided significance level. These calculations should be interpreted descriptively because observed post hoc power is a function of the estimated effect size and event count.

Table 6: Initial Observed Power by Overall Group and Subgroup

Analysis	Subgroup	n	Events	Observed HR	Event Rate	Observed Power	Interpretation
Whole Group	All	1151	96	0.517	0.083	0.839	At or above 80%
Sex	Female	951	81	0.459	0.085	0.897	At or above 80%
Sex	Male	200	15	1.101	0.075	0.020	Below 80%
Race/Ethnicity	White	596	51	0.539	0.086	0.486	Below 80%
Race/Ethnicity	Black	327	21	0.287	0.064	0.732	Below 80%
Race/Ethnicity	Hispanic	203	20	0.628	0.099	0.115	Below 80%
Race/Ethnicity	Asian/PI	14	3	0.068	0.214	0.535	Below 80%
Race/Ethnicity	Other	11	1	237.311	0.091	0.689	Below 80%

The initial power estimates indicate that the overall study and some larger subgroups had adequate power to detect the observed treatment effect. However, sparse event counts limited estimates for smaller subgroups. In particular, hazard ratios were estimated from very few events for male, Asian/Pacific Islander, and Other races, making the calculated power unstable and primarily descriptive.

5.6 Power Analysis: Required Sample Sizes for a Future Study

The subgroup analyses revealed substantial imbalance in sample sizes across sex and race/ethnicity groups, leaving several subgroups severely underpowered. This section uses the observed effect sizes (HRs) from the stratum-specific models as assumed true effects in a prospective power analysis. We estimate the sample sizes a future study would need to detect these effects.

For subgroups with unstable HR estimates (Asian/PI, Other, Male), the analysis uses a range of plausible HRs to illustrate the needed sample sizes to detect effects of various magnitudes.

5.6.1 Reliable Subgroups

These subgroups had sufficient events for stable HR estimation.

Table 7: Required Sample Sizes by Subgroup and Power Level ($\alpha = 0.025$)

Subgroup	Observed HR	Obs. n	Obs. Events	Event Rate	n (80%)	n (85%)	n (90%)	n (95%)
Whole Group	0.517	1151	96	0.083	264	300	348	420
Sex: Female	0.459	951	81	0.085	188	212	247	294
Race: White	0.539	596	51	0.086	293	339	386	468
Race: Black	0.287	327	21	0.064	109	109	125	156
Race: Hispanic	0.628	203	20	0.099	447	508	589	711

5.6.2 Underpowered Subgroups — Plausible HR Range

For the male, Asian/PI, and Other subgroups, the observed HRs were unreliable due to few event counts. Instead, Table 8 shows required sample sizes across a range of plausible HRs (0.50 to 0.90), using each subgroup's observed event rate.

Table 8: Required Sample Sizes for 80% Power Across Assumed Hazard Ratios

Subgroup	Assumed HR	Observed Event Rate	Required n for 80% Power
Sex: Male	0.5	0.075	267
Sex: Male	0.6	0.075	494
Sex: Male	0.7	0.075	1000
Sex: Male	0.8	0.075	2547
Sex: Male	0.9	0.075	11427
Race: Asian/PI	0.5	0.214	94
Race: Asian/PI	0.6	0.214	173
Race: Asian/PI	0.7	0.214	350
Race: Asian/PI	0.8	0.214	892
Race: Asian/PI	0.9	0.214	4000
Race: Other	0.5	0.091	220
Race: Other	0.6	0.091	407
Race: Other	0.7	0.091	825
Race: Other	0.8	0.091	2101
Race: Other	0.9	0.091	9427

5.6.3 Events Required vs. Sample Size

Table 10 shows the required number of events from the required total sample size for the reliable subgroups at 80% power.

Table 9: Required Events and Sample Sizes for 80% Power

Subgroup	Observed HR	Event Rate	Events Required for 80% Power	Required n for 80% Power
Whole Group	0.517	0.083	22	264
Sex: Female	0.459	0.085	16	188
Race: White	0.539	0.086	25	293
Race: Black	0.287	0.064	7	109
Race: Hispanic	0.628	0.099	44	447

5.6.4 Power Curve Plot

5.6.5 Sample Sizes Required for 80% Power

Table 10: Adequacy of ACTG 320 Sample Sizes for 80% Power

Subgroup	Observed HR	ACTG 320 n	Required n for 80% Power	ACTG 320 Status
Whole Group	0.517	1151	264	Adequate
Sex: Female	0.459	951	188	Adequate
Race: White	0.539	596	293	Adequate
Race: Black	0.287	327	109	Adequate
Race: Hispanic	0.628	203	447	Underpowered

6 Discussion

This paper reanalyzed the ACTG 320 clinical trial [Hammer1997] to evaluate how demographic subgroup representation and observed event counts affect statistical power in time-to-event analyses. We successfully replicated the primary treatment effect: indinavir-containing combination therapy reduced the hazard of progression to AIDS or death by approximately 48% relative to the control regimen ($HR = 0.517$), consistent with the original findings. This

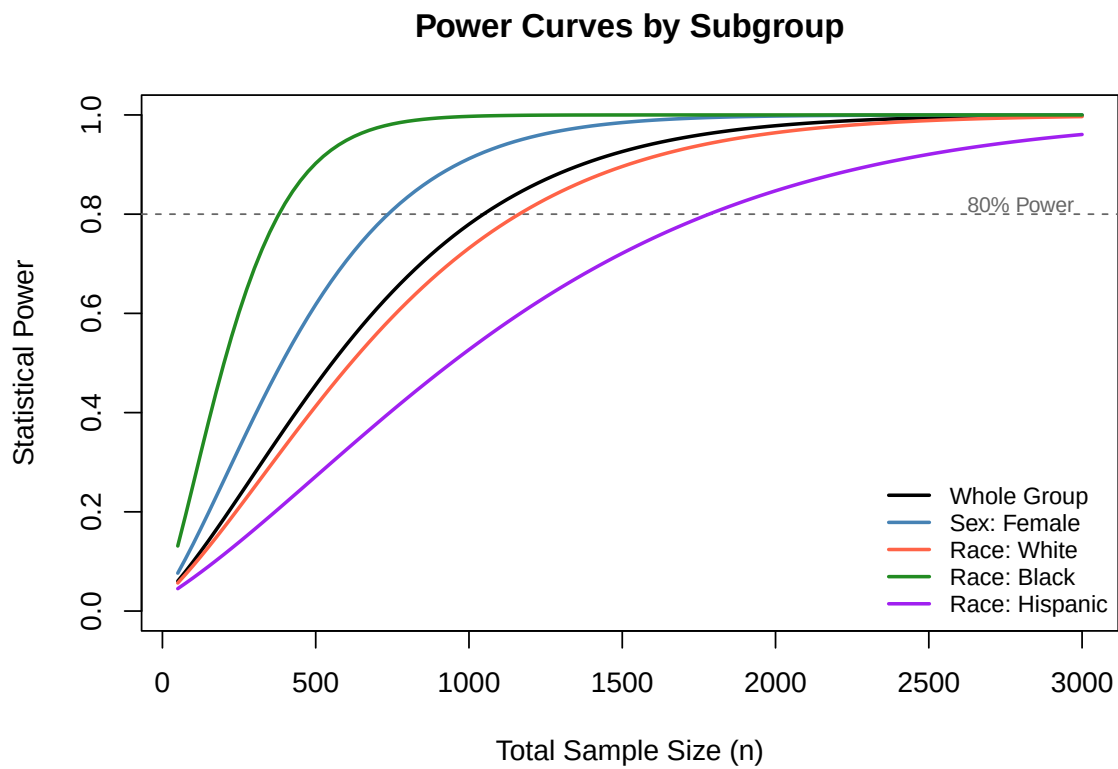


Figure 2: Power curves showing achieved power as a function of sample size for each reliable subgroup, with the 80% power threshold shown.

result was stable across all baseline model specifications and was statistically significant (log-rank $p = 0.0012$), confirming that the overall study had adequate power for its intended purpose.

Subgroup analyses revealed a more complicated picture. We did not detect statistically significant treatment-by-sex or treatment-by-race/ethnicity interaction effects. The data do not provide evidence that indinavir’s benefit differed across demographic groups. However, this result is not evidence of homogeneous treatment effects. The power analysis demonstrates that most subgroups were never in a position to detect meaningful differences. The male stratum contributed only 15 events across 200 enrolled participants, leaving the stratum-specific Cox model severely underpowered and its hazard ratio estimate ($HR = 1.101$) unreliable. The Asian/Pacific Islander and Other race subgroups contributed 3 and 1 events, respectively, making their estimates entirely uninformative. Even among the more stable subgroups, only the overall group and the female subgroup exceeded the 80% power threshold at the observed effect sizes. The Hispanic subgroup enrolled 203 participants but remained underpowered because its low event count (20 events) was insufficient to detect a moderate treatment effect. It would require enrollment of approximately 447 participants to achieve 80% power.

The central lesson of the power analysis is that observed events, not enrollment, determine power in survival studies. This distinction is not always apparent in the clinical trial literature. Representation is often discussed in terms of enrollment proportions rather than event-driven inferential capacity. A subgroup that is nominally represented in a trial may still contribute too few events to support any meaningful conclusion about treatment heterogeneity. The results here quantify that gap explicitly: for the underpowered subgroups, achieving 80% power would require enrollment increases ranging from modest (approximately 94 participants for Asian/PI assuming $HR = 0.50$) to very large (over 11,000 for male participants assuming $HR = 0.90$). These figures illustrate that when the true treatment effect is small or moderate, the sample size requirements for subgroup inference can be prohibitive within a single trial.

6.1 Limitations

This analysis had several limitations. First, the analysis relies on secondary use of a publicly available dataset and can only use variables included in the public data release. Secondary endpoints from the original trial, CD4 cell count trajectories, and HIV-1 RNA concentrations, are not available, limiting this reanalysis to the primary time-to-event outcome. Second, the total number of primary endpoint events was small (96 across 1,151 participants), which constrains the precision of all subgroup analyses regardless of modeling approach. Third, the power calculations rely on asymptotic approximations under the Cox proportional hazards framework and assume that hazard ratios remain proportional over the follow-up period. This assumption may not hold in all subgroups, given the sparse event counts. Fourth, the race/ethnicity variable in the available dataset contains only five levels; the American Indian/Alaska Native category described in the original trial protocol is absent from the public data. The ethnicity

analysis has less coverage than the original design intended. Finally, post-hoc power estimates for the most underpowered subgroups, male, Asian/PI, and Other, should be treated as descriptive only because the underlying hazard ratio estimates driving those calculations are themselves unstable.

6.2 Implications and Future Research

The findings of this paper have direct implications for the design of future clinical trials where subgroup-specific inference is a scientific objective. Trials powered only for overall treatment comparisons are structurally unable to detect differential treatment responses in underrepresented demographic groups, not because those differences do not exist, but because the study lacks the power to detect them. This problem is particularly consequential for historically underrepresented populations in clinical research, who may face distinct biological, pharmacological, or social determinants of treatment response. As @Khan2020 and @Poon2013 document, this pattern is not specific to HIV trials; it is a persistent feature of pivotal drug development across therapeutic areas.

Practical solutions exist but require intentional trial design. Prospective enrollment targets for key demographic subgroups, set based on anticipated event rates and minimum detectable effect sizes rather than proportional representation alone, would ensure that subgroup analyses are interpretable rather than merely exploratory. Extending follow-up duration increases event accrual without requiring additional enrollment, which may be a cost-effective strategy to improve subgroup power in trials with low overall event rates. Another option is adaptive event-driven designs that trigger subgroup analyses only after reaching subgroup-specific event thresholds. For situations where no single trial can achieve adequate subgroup enrollment, Bayesian hierarchical models that borrow information across subgroups under assumptions of partial exchangeability, and pooled meta-analytic frameworks across multiple trials, may provide the inferential capacity that individual studies cannot. Future methodological work should evaluate the performance of these approaches specifically under the sparse-event subgroup conditions illustrated here.

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